



SA-RAS: Speaker-Aware Style Retrieval Augmented Generation for Expressive Zero-Shot Text-to-Speech Synthesis

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Abstract

Zero-shot text-to-speech (TTS) models can clone an unseen speaker by reference speech. Given multiple reference speeches, Retrieval Augmented Generation (RAG) shows potential in enhancing synthesis quality. The existing method faces two key limitations: semantically-biased retrieval tends to introduce style mismatch, and the deterministic mapping between text and speech fails to fully capture the unique style of the speaker. To solve these, we propose a Speaker-Aware Retrieval Augmented TTS Synthesis (SA-RAS) model. It consists of a two-stage TTS model, Split-TTS and a zero-shot CLAP. The former embeds style and timbre separately to obtain a style space for retrieval. The latter optimizes the correlation between text and style by incorporating speaker information into the retrieval model. Experiments prove that SA-RAS boosts the style expressiveness of synthesized speech. Speech samples are available at the following link <https://zuoguaidong.github.io/2025interspeech/>.

Index Terms: Zero-shot TTS, retrieval augmentation, contrastive learning

1. Introduction

Zero-shot text-to-speech (TTS) is a powerful synthesis model [1] [2]. It can extract features such as prosody and speaker features from reference speech to generate natural speech for unseen speakers [3] [4]. In practical applications, it is quite common to have multiple speeches of the target speaker available as references. The key to fully leveraging the value of these data lies in precisely selecting the most fitting reference speech for TTS according to the target synthesized text. This process is of great significance for enhancing the style expressiveness of the synthesized speech [5].

Retrieval-Augmented Generation (RAG) has achieved remarkable success in optimizing the generation performance of large language models (LLMs) [6] [7]. In light of this success, RAG has also started to penetrate into audio tasks and is applied in audio deepfake detection [8], audio captioning [9], and speech synthesis [10]. Xue et al. [10] used Context-Aware Contrastive Language-Audio Pretraining (CA-CLAP) as the retrieval method to select text and audio prompts. This optimization of speech prompts effectively enhanced the synthesis performance of the prompt-based TTS system. However, there is still a phenomenon of style retrieval bias, which is mainly attributable to the following two causes.

Firstly, both speech and text contain rich semantic information, and different audio encoders can extract various features, such as content, prosody, and timbre [11]. With Contrastive

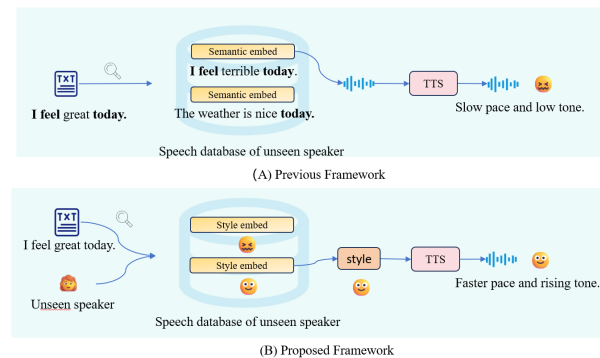


Figure 1: Comparison between the previous and the proposed RAG-enhanced zero-shot TTS frameworks

Language-Audio Pretraining (CLAP) [12] being trained to learn the correspondence between speech and text for retrieval, the embeddings of speech and text are mapped into a shared latent space. This embedding-mapping process, in turn, guides the audio encoder in CLAP to tend to extract semantic features. However, there are cases of high semantic similarity but low style similarity. Directly establishing a retrieval relationship in the semantic space is highly likely to result in retrieving audio with a mismatched style, as shown in Figure 1(A).

Secondly, there exists a one-to-many relationship between text and styles. Different speakers may exhibit style differences when expressing the same text [13]. If retrieval is conducted based on the deterministic mapping between text and speech, it becomes impossible to capture and represent the distinct style characteristics of each individual speaker, let alone those of unseen speakers. Consequently, further improving the retrieval model to adapt to zero-shot TTS is needed.

For the first issue, it is the text embedding that should be mapped into the style space. Speaking style and speaker timbre are highly entangled as both of them affect the prosody patterns of speech. However, timbre, tied to a speaker's unique physiology, doesn't relate directly to text-conveyed style and can disrupt style retrieval. Therefore, it's necessary to construct a TTS framework that can split timbre and style, so as to obtain a style encoder that can extract timbre-independent style feature for the retrieval model. In regard to the second issue, inspired by zero-shot [14], we introduce speaker information into the retrieval model. It serves to direct the model to dynamically adjust the retrieval relationship between text and style, addressing the one-to-many relationship between text and styles caused by speakers. Moreover, it enables the model to maintain the abil-

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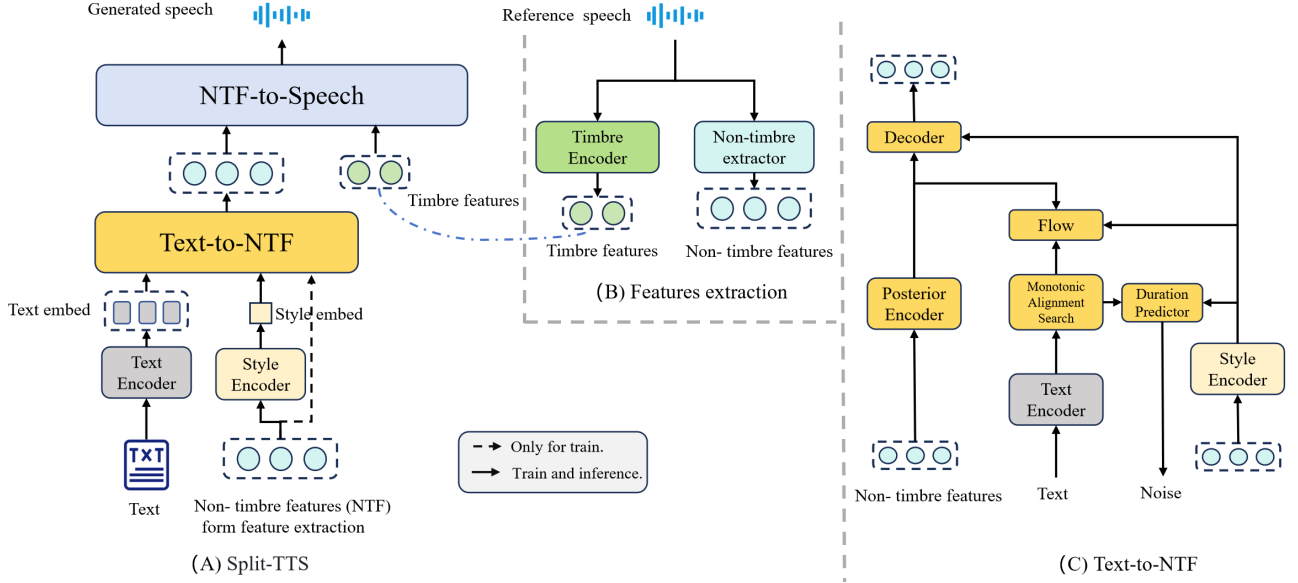


Figure 2: (A) An overview of our proposed Split-TTS (B) The features extraction process (C) The details of the Text-to-NTF Model

ity to conduct personalized retrieval even for unseen speakers. The enhancements in these two facets pave the way for the successful transfer of the RAG method, which is usually utilized in strong semantic-matching tasks, to the zero-shot TTS task.

Therefore, as shown in Figure 1(B), we propose the Speaker-Aware Retrieval-Augmented Synthesis (SA-RAS) model. We propose Split-TTS, a two-stage TTS framework, to obtain a style encoder capable of extracting style feature independent of timbre and semantics for the text to style retrieval. The style and timbre of the reference speech are respectively embedded in two stages, ensuring a retrieval-compatible style encoder.

To improve the retrieval model, we propose the Zero-shot Contrastive Language-Audio Pre-training (Zero-shot CLAP) model. With the aid of speaker embedding, Zero-shot CLAP is trained to learn the correspondence between texts and styles of different speakers, extracting appropriate style-related features from the text for retrieval. By this method, it breaks through the limitations of deterministic mapping.

Overall, our paper makes the following contributions:

- We propose Zero-shot CLAP and innovatively incorporate speaker embedding into the retrieval model.
- To achieve a retrieval between text and style, we design a two-stage Split-TTS, which separates style and non-style features, to obtain a retrieval-compatible style encoder.
- We conduct a comprehensive evaluation on the LibriTTS [15] and ESD [16] datasets, demonstrating that SA-RAS improves the style expressiveness of zero-shot TTS with multiple references.

2. Methodology

The SA-RAS consists of two parts: the style retrieval model Zero-shot CLAP and the speech generation model Split-TTS. These two models assist each other. Zero-shot CLAP uses the style encoder well-trained in Split-TTS to train a semantic style encoder for style retrieval. Split-TTS, on the other hand, leverages the retrieval ability of Zero-shot CLAP to augment the ex-

pressiveness of the synthesized speech.

2.1. Split-TTS Model

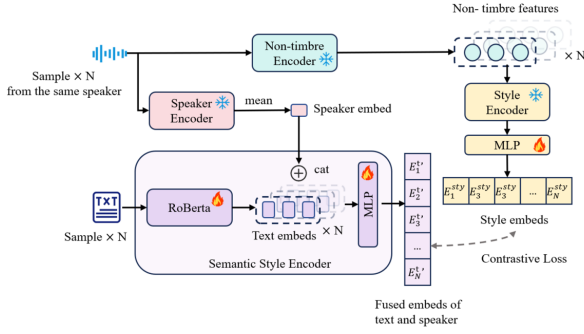
The proposed Split-TTS, a two-stage zero-shot TTS model, is trained to provide a style encoder and perform speech synthesis for SA-RAS. Split-TTS is capable of splitting and separately embedding style and timbre. An overview of the Split-TTS framework is presented in Figure 2(A). The first stage is a generation process from text to non-timbre features (NTF), and the second stage is from NTF to speech synthesis.

Before training the model, feature extraction is required to obtain the NTF and timbre features needed for the two-stage training of Split-TTS, as shown in Figure 2(B). Seed-VC[17], which has the main architecture of a diffusion transformer and a length regulator, has powerful voice conversion capability. We use Seed-VC to process the data: fixing one speech as reference to extract NTF with a unified timbre. By doing so, the features are completely stripped of effective timbre information. Since the second-stage model is based on Seed-VC, we extract timbre features in Seed-VC for timbre embedding.

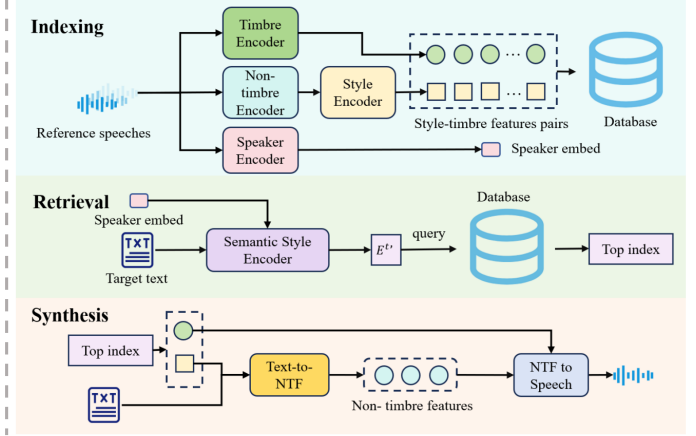
The text-to-NTF model is based on VITS [18], which adopts the architecture of conditional variational autoencoder (CVAE), significantly enhancing the naturalness of speech generation. The architectural details are shown in Figure 2(C). We introduce the GST [19] as the style encoder to extract global style embedding from the NTF. The extracted global style embedding is then added to the input of the duration predictor, normalizing flow, posterior encoder, and decoder. In the second stage, we design the NTF-to-speech model based on Seed-VC. We perform adaptive fine-tuning on the diffusion transformer to recover speech from timbre features and the NTF generated in the first stage.

2.2. Zero-shot CLAP

Inspired by CA-CLAP [10], we employ CLAP [12] to train a semantic style encoder for multi-modal retrieval in style space. The overall architecture is shown in Figure 3(A).



(A) The architecture of Zero-shot TTS.



(B) The application the SA-RAS.

Figure 3: An overview of our proposed Zero-shot CLAP and the application of the SA-RAS

Our model utilizes three encoders to separately process the audio data X_i^a and text data X_i^t , where (X_i^a, X_i^t) is one of the audio-text pairs indexed by i . Specifically, the style embedding E_i^{sty} and the speaker embedding E_i^{spk} are respectively obtained from the style encoder $f_{\text{sty}}(\cdot)$ trained in Split-TTS and the pre-trained speaker encoder H/ASP [20] $f_{\text{spk}}(\cdot)$. We sum and average all the E_i^{spk} of the same speaker to obtain the speaker embedding E^{spk} . The text encoder RoBERTa [21] $f_{\text{text}}(\cdot)$ generates the text embedding E_i^t . $E_i^{t'}$ is obtained by concatenating E^{spk} and E_i^t to achieve information fusion.

$$E_i^{\text{sty}} = \text{MLP}_{\text{audio}}(f_{\text{sty}}(X_i^a)) \quad (1)$$

$$E^{\text{spk}} = \frac{1}{K} \sum_{i=0}^K f_{\text{spk}}(X_i^a) \quad (2)$$

$$E_i^{t'} = \text{MLP}_{\text{text}}(f_{\text{text}}(X_i^t) \text{ cat } E^{\text{spk}}) \quad (3)$$

K refers to the number of speech utterances owned by a certain speaker. The multi-layer perceptron (MLP) serves as a projection layer, which maps $E_i^{t'}$ and E_i^{sty} to the same dimension.

During the training process, we use N pairs of (X_i^a, X_i^t) from the same speaker, along with the speaker's E_i^{spk} as inputs, and feed them into the Zero-shot CLAP encoder. This results in N pairs of $(E_i^{t'}, E_i^{\text{sty}})$. Subsequently, we compute the cosine similarity for every pair among these pairs, thus generating an $N \times N$ matrix. Among them, the N values on the diagonal of the matrix correspond to the cosine similarities of the correctly matched pairs in each batch. We follow the same contrastive learning loss as that in [12].

$$L = \frac{1}{2N} \sum_{i=1}^N \left(\log \frac{\exp(E_i^{\text{sty}} \cdot E_i^{t'} / \tau)}{\sum_{j=1}^N \exp(E_i^{\text{sty}} \cdot E_j^{t'} / \tau)} + \log \frac{\exp(E_i^{t'} \cdot E_i^{\text{sty}} / \tau)}{\sum_{j=1}^N \exp(E_i^{t'} \cdot E_j^{\text{sty}} / \tau)} \right) \quad (4)$$

The loss aims to maximize the cosine similarity values corresponding to the N correct pairs $(E_i^{\text{sty}} \cdot E_i^{t'})$ while minimizing the remaining $N^2 - N$ incorrect pairs.

Only the text encoder and the MLP layers are optimized in the training. As a result, neither the style embedding space obtained from the training of Split-TTS nor the speaker embedding space is affected.

2.3. Inference stage

As shown in Figure 3(B), the application of the SA-RAS includes three steps: indexing, retrieval, and synthesis.

- **Indexing:** We utilize the style encoder and timbre encoder to encode all the available reference speeches into style embedding and timbre features constructing a database. And we compute the average speaker embedding from all candidate speeches for CA-CLAP.
- **Retrieval:** The input text is encoded and fused with the speaker embedding to generate the retrieval query. The style embeddings in the database are then projected through the MLP layer, and similarity scores are calculated between the query and each embedding to select the top one.
- **Synthesis:** With the retrieved style embedding and timbre features from the selected reference speech, Split-TTS generates the target speech.

3. Experiment

3.1. Datasets

We use the LibriTTS [15] and ESD [16] datasets in the experiment and adjust the sampling rate to 22050Hz.

LibriTTS is a multi-speaker English corpus that contains approximately 585 hours of speech from over 2300 speakers. We merge all the training and validation sets of LibriTTS and carry out a random re-splitting according to speakers. The validation set and the test set contain all the data from 10 and 12 speakers respectively. This data partitioning method is applicable to both the Zero-shot CLAP and Split TTS models.

ESD is an Emotional Speech Database that consists of 350 parallel utterances spoken by 10 native English and 10 native Chinese speakers and covers 5 emotion categories. We employ this dataset to evaluate the retrieval space.

Table 1: Objective and subjective experiments comparing TTS performance across various retrieval methods. We evaluate the 95% confidence intervals for NMOS, S-SMOS and ST-MOS.

model	MCD↓	Energy↓	F0↓	EECS↑	NMOS↑	S - SMOS↑	ST-MOS↑
Self	6.003	6.710	20.036	0.896	4.046±0.133	3.980±0.147	3.913±0.145
CA - CLAP	<u>6.678</u>	7.935	22.688	0.864	3.408±0.122	3.117±0.143	<u>3.189±0.134</u>
Random	7.022	8.648	24.186	0.840	2.347±0.155	2.964±0.148	2.934±0.164
w/o Speaker	6.670	6.935	22.183	0.873	3.505±0.121	3.168±0.139	3.124±0.145
Proposed	6.860	<u>7.494</u>	21.559	<u>0.872</u>	3.827±0.135	3.556±0.143	3.653±0.136

3.2. Comparison Experiments

Our work focuses on enhancing the style expressiveness of zero-shot TTS through retrieval. Thus, we conduct comparative experiments of different retrieval methods on Split-TTS, including the speech paired with the target text, CA-CLAP [10], the randomly selected speech, and Zero-shot CLAP.

For CA-Clap, The model and hyperparameters are kept consistent with those in [10]. Since context is not considered in the experiment, we remove the context-related modules in the model for comparison.

We train the first stage of Split-TTS for 100k steps on one NVIDIA RTX 2080 GPU with a batch size of 32. For both Zero-shot CLAP and CA-CLAP, 32 epochs of training are performed with a batch size of 32.

The comparison experiment is conducted on unseen speakers. To ensure there are sufficient samples to support the retrieval, apart from the speech that matches the currently synthesized text, all the other speeches are used as samples in the style database for selection.

3.3. Objective Evaluation

Following the experiment of CA-CLAP, We utilize three metrics including energy, F0, and mel-cepstral distortion (MCD) aligned by Dynamic Time Warping (DTW). We also evaluate the Emotion Cosine Similarity (EECS) between synthesized speech and ground-truth speech using emotion2vec [22], which serves to assess the similarity of style from an emotional perspective.

The objective evaluation results are presented in Table 1. Comprehensively considering multiple metrics, the results show that 1) Using the speech that is paired with the text for reconstruction yields the best results. 2) Zero-shot CLAP comprehensively demonstrates a better performance than CA-CLAP, further enhancing the style expressiveness of TTS. 3) Random selection leads to the worst results, which shows that retrieval augmentation is of great significance in TTS.

Speeches conveying the same emotion possess certain similarities in style. Therefore, from the ESD dataset, we select speeches where the same speaker expressed identical text with happy and sad emotions respectively. Then embeddings are extracted from the audio encoders used in Zero-shot CLAP and CA-CLAP to study their embedding space. We use t-SNE [23] to visualize the results. As shown in Figure 4, the result shows that the audio encoder’s embedding distribution of CA-CLAP is uniform across emotions, while Zero-shot CLAP can better distinguish between different emotions, which demonstrates the superiority of Zero-shot CLAP in style retrieval.

3.4. Subjective evaluation

We conduct three Mean Opinion Score (MOS) subjective tests. The Naturalness MOS (NMOS) is used to evaluate the natural-

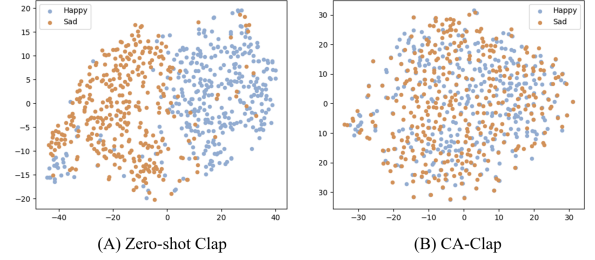


Figure 4: The visualization of the speech embeddings from different retrieval models

ness and quality. The Style Similarity MOS (S-SMOS) is employed to compare the style similarity between the ground-truth speech and the synthesized speech. The Style and Text compatibility MOS (ST-MOS) is carried out to assess the degree of compatibility between the style of the synthesized speech and the text.

We invite 15 native listeners to rate 15 samples on a scale from 1 to 5. The samples are selected from three unseen speakers with different speaking styles and are relatively representative. The results of the subjective evaluation are presented in Table 1. The results show that our proposed model outperforms all the comparative models, which demonstrates its powerful retrieval capabilities.

3.5. Ablation Study

To study the impact of incorporating speaker embedding into the retrieval model, we conducted an ablation experiment. We removed the speaker encoder from Zero-shot CLAP and then carried out training with the same settings. The results shown in Table 1 indicate that 1) style-based retrieval after ablation is still better than semantics-based retrieval. 2) In subjective evaluation, incorporating speaker information can boost the style expressiveness of the synthesized speech.

4. Conclusion

In this paper, we propose SA-RAS, a speaker-aware style retrieval-augmented zero-shot TTS architecture tailored for multiple reference-speech scenarios. The proposed model is composed of Split-TTS and Zero-shot CLAP. Split-TTS obtains a style encoder for both TTS and the retrieval model. And the Zero-shot CLAP is a speaker-aware style retrieval model. It enables the optimal selection of reference speeches by effectively leveraging the RAG paradigm in zero-shot TTS. Evaluations conducted on the LibriTTS and ESD datasets demonstrate that the SA-RAS model enhances the style expressiveness of the synthesized speech.

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